YEAR 6

HOMESCHOOL MATHEMATICS WORKBOOK

A 12-week guided program | Australian Year 6 level

Six units • worked examples • 144 practice questions • 24 review questions • complete parent answer key

Student name: _____

Start date: _____

Parent / tutor: _____

Print one copy for each student. Keep the answer key at the back for parent use. Show working, not just answers.

How to use this workbook

Study 5 days per week for about 45-60 minutes. Each unit is designed for two weeks. Spend the first days learning and practising, then complete the review quiz independently.

Suggested lesson: 10 min warm-up; 15 min teaching; 10 min guided example; 20 min worksheet; 5 min marking. For the 24-question worksheet, complete 6 questions per practice day.

Target: 80% or higher on the review quiz. If a skill is difficult, revisit the worked example and try new similar questions before moving on.

12-week teaching plan

Weeks 1-2: Unit 1 - Whole Numbers, Factors and Order of Operations

Weeks 3-4: Unit 2 - Fractions, Decimals and Percentages

Weeks 5-6: Unit 3 - Patterns, Algebra and Equations

Weeks 7-8: Unit 4 - Measurement, Area, Perimeter and Volume

Weeks 9-10: Unit 5 - Geometry, Angles and Coordinates

Weeks 11-12: Unit 6 - Data, Graphs and Probability

Progress record - Student 1

Unit 1: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 2: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 3: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 4: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 5: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 6: Date _____ Quiz score ___ / 4 Parent initials _____

Progress record - Student 2

Unit 1: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 2: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 3: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 4: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 5: Date _____ Quiz score ___ / 4 Parent initials _____

Unit 6: Date _____ Quiz score ___ / 4 Parent initials _____

UNIT 1 | Whole Numbers, Factors and Order of Operations

Learning goal

Read large numbers, calculate accurately, use factors and multiples, and follow order of operations.

Teacher notes

Explain the rule aloud, model the first question, then ask the student to explain each step back to you. Use paper, a ruler or household objects as appropriate.

Worked examples

Example 1: Calculate 24 + 6 x 3.

Multiply first: 6 x 3 = 18, then 24 + 18 = 42.

Example 2: Find the highest common factor of 18 and 30.

Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. HCF = 6.

Independent worksheet | 24 questions

Show your working. You may use the back of the page for extra space.

1. 3,468 + 2,579

2. 7,850 + 1,645

3. 12,947 + 8,065

4. 37,562 + 28,479

5. 8,000 - 3,476

6. 14,005 - 7,869

7. 52,030 - 18,875

8. 100,000 - 65,784

UNIT 1 | Worksheet continued

9. 48 x 7

10. 136 x 6

11. 24 x 15

12. 208 x 14

13. 864 / 8

14. 945 / 9

15. 672 / 12

16. 1512 / 14

UNIT 1 | Worksheet continued

17. 8 + 5 x 4

18. 32 + 6 x 3

19. 15 + 7 x 8

20. 45 + 4 x 9

21. (8 + 5) x 4

22. (12 + 9) x 3

23. (17 + 8) x 6

24. (24 + 6) x 5

UNIT 1 | Review quiz

Complete without help. 1 mark per question. Score: _____ / 4

1. Find the factors of 24.

2. Find the first five multiples of 7.

3. Round 68,749 to the nearest thousand.

4. A school buys 18 packs of 24 pencils. How many pencils?

What I found easy: _____

What I need to practise: _____

UNIT 2 | Fractions, Decimals and Percentages

Learning goal

Compare, convert and calculate with fractions, decimals and percentages.

Teacher notes

Explain the rule aloud, model the first question, then ask the student to explain each step back to you. Use paper, a ruler or household objects as appropriate.

Worked examples

Example 1: Calculate 3/4 + 1/8.

Convert 3/4 to 6/8; then 6/8 + 1/8 = 7/8.

Example 2: Find 25% of $80.

25% means one quarter; 80 / 4 = $20.

Independent worksheet | 24 questions

Show your working. You may use the back of the page for extra space.

1. Simplify 6/12

2. Simplify 12/18

3. Simplify 15/25

4. Simplify 21/28

5. 1/4 + 2/4

6. 2/5 + 1/5

7. 3/8 + 1/8

8. 5/6 - 1/6

UNIT 2 | Worksheet continued

9. 1/2 + 1/4

10. 2/3 + 1/6

11. 3/4 - 1/8

12. 5/6 - 1/3

13. 0.6 + 0.25

14. 3.7 + 2.85

15. 8.4 - 3.75

16. 2.5 x 4

UNIT 2 | Worksheet continued

17. Convert 1/2 to a decimal

18. Convert 3/4 to a decimal

19. Convert 0.2 to a percentage

20. Convert 0.35 to a percentage

21. Find 10% of 250

22. Find 25% of 120

23. Find 50% of 86

24. Find 20% of 75

UNIT 2 | Review quiz

Complete without help. 1 mark per question. Score: _____ / 4

1. A $60 shirt is 25% off. Sale price?

2. Which is larger: 0.7 or 2/3?

3. Write 45% as a simplified fraction.

4. A recipe needs 3/4 cup of milk; you have 1/2 cup. How much more?

What I found easy: _____

What I need to practise: _____

UNIT 3 | Patterns, Algebra and Equations

Learning goal

Identify rules, use variables and solve simple one-step equations.

Teacher notes

Explain the rule aloud, model the first question, then ask the student to explain each step back to you. Use paper, a ruler or household objects as appropriate.

Worked examples

Example 1: Find the next number: 4, 9, 14, 19, ...

Add 5 each time, so the next number is 24.

Example 2: Solve x + 7 = 19.

Subtract 7 from both sides: x = 12.

Independent worksheet | 24 questions

Show your working. You may use the back of the page for extra space.

1. Next: 3, 6, 9, 12, ...

2. Next: 7, 12, 17, 22, ...

3. Next: 2, 4, 8, 16, ...

4. Next: 100, 90, 80, 70, ...

5. Rule for 5, 10, 15, 20

6. Rule for 2, 6, 10, 14

7. Rule for 40, 35, 30, 25

8. Rule for 3, 9, 27, 81

UNIT 3 | Worksheet continued

9. Find 2n when n = 6

10. Find n + 8 when n = 13

11. Find 3n + 1 when n = 4

12. Find 20 - n when n = 7

13. Solve x + 5 = 14

14. Solve x + 12 = 30

15. Solve x - 6 = 17

16. Solve x - 9 = 8

UNIT 3 | Worksheet continued

17. Solve 4x = 28

18. Solve 6x = 54

19. Solve x / 5 = 7

20. Solve x / 3 = 12

21. Find the 10th term of 3, 6, 9, ...

22. Find the 8th term of 4, 8, 12, ...

23. Find 5n - 2 when n = 3

24. Solve x + 18 = 41

UNIT 3 | Review quiz

Complete without help. 1 mark per question. Score: _____ / 4

1. A number doubled is 46. Find the number.

2. Next two: 6, 11, 16, 21, ...

3. Write an expression for 7 more than a number n.

4. If 3 identical pens cost $15, what is one pen?

What I found easy: _____

What I need to practise: _____

UNIT 4 | Measurement, Area, Perimeter and Volume

Learning goal

Convert metric units and calculate perimeter, area and volume in practical settings.

Teacher notes

Explain the rule aloud, model the first question, then ask the student to explain each step back to you. Use paper, a ruler or household objects as appropriate.

Worked examples

Example 1: Find the area of a rectangle 8 cm long and 5 cm wide.

Area = length x width = 8 x 5 = 40 square cm.

Example 2: Find the volume of a box 4 cm x 3 cm x 2 cm.

Volume = length x width x height = 4 x 3 x 2 = 24 cubic cm.

Independent worksheet | 24 questions

Show your working. You may use the back of the page for extra space.

1. Convert 3 m to cm

2. Convert 250 cm to m

3. Convert 4 km to m

4. Convert 1500 m to km

5. Convert 2 L to mL

6. Convert 750 mL to L

7. Convert 3 kg to g

8. Convert 1250 g to kg

UNIT 4 | Worksheet continued

9. Perimeter: rectangle 6 cm x 4 cm

10. Perimeter: rectangle 12 m x 7 m

11. Perimeter: square side 9 cm

12. Perimeter: square side 15 m

13. Area: rectangle 7 cm x 6 cm

14. Area: rectangle 14 m x 3 m

15. Area: square side 8 cm

16. Area: square side 11 m

UNIT 4 | Worksheet continued

17. Volume: 2 x 3 x 5 cm box

18. Volume: 4 x 4 x 4 cm cube

19. Volume: 10 x 2 x 3 cm box

20. Volume: 6 x 5 x 2 cm box

21. Convert 90 minutes to hours and minutes

22. Convert 2 h 15 min to minutes

23. Elapsed time: 9:20 am to 11:05 am

24. Elapsed time: 1:45 pm to 3:10 pm

UNIT 4 | Review quiz

Complete without help. 1 mark per question. Score: _____ / 4

1. A garden is 5 m by 4 m. How much fencing around it?

2. How many square metres of lawn cover that garden?

3. A tank holds 2.5 L. How many millilitres?

4. A rectangular box measures 6 cm x 5 cm x 4 cm. Volume?

What I found easy: _____

What I need to practise: _____

UNIT 5 | Geometry, Angles and Coordinates

Learning goal

Classify angles, reason about shapes and locate points on a coordinate plane.

Teacher notes

Explain the rule aloud, model the first question, then ask the student to explain each step back to you. Use paper, a ruler or household objects as appropriate.

Worked examples

Example 1: Two angles on a straight line total 180 degrees. One is 65 degrees.

Missing angle = 180 - 65 = 115 degrees.

Example 2: A triangle has angles 45 and 70 degrees. Find the third.

Triangle angles total 180; 180 - 45 - 70 = 65 degrees.

Independent worksheet | 24 questions

Show your working. You may use the back of the page for extra space.

1. Classify 35 degrees

2. Classify 90 degrees

3. Classify 130 degrees

4. Classify 180 degrees

5. Missing angle on line: 70 degrees

6. Missing angle on line: 125 degrees

7. Missing angle on line: 48 degrees

8. Missing angle on line: 90 degrees

UNIT 5 | Worksheet continued

9. Triangle angles: 60, 60, ?

10. Triangle angles: 45, 75, ?

11. Triangle angles: 30, 90, ?

12. Triangle angles: 80, 40, ?

13. How many sides has a hexagon?

14. How many sides has an octagon?

15. How many degrees in a full turn?

16. How many degrees in a quarter turn?

UNIT 5 | Worksheet continued

17. Quadrant of (3, 4)

18. Quadrant of (-3, 4)

19. Quadrant of (-3, -4)

20. Quadrant of (3, -4)

21. Reflect (2, 5) in x-axis

22. Reflect (4, -1) in y-axis

23. Translate (1, 2) right 3

24. Translate (-2, 3) down 4

UNIT 5 | Review quiz

Complete without help. 1 mark per question. Score: _____ / 4

1. Triangle angles are 50 and 55 degrees. Find third.

2. How many lines of symmetry does a square have?

3. What is the coordinate of the origin?

4. A half turn equals how many degrees?

What I found easy: _____

What I need to practise: _____

UNIT 6 | Data, Graphs and Probability

Learning goal

Read and describe data, calculate averages, and compare chance with experiment.

Teacher notes

Explain the rule aloud, model the first question, then ask the student to explain each step back to you. Use paper, a ruler or household objects as appropriate.

Worked examples

Example 1: Find the mean of 4, 6, 8, 10.

Add 4 + 6 + 8 + 10 = 28; divide by 4 = 7.

Example 2: A fair die has six equally likely faces. What is P(rolling a 2)?

One favourable outcome out of six: 1/6.

Independent worksheet | 24 questions

Show your working. You may use the back of the page for extra space.

1. Mean of 2, 4, 6

2. Mean of 5, 7, 9

3. Mean of 3, 6, 9, 12

4. Mean of 10, 12, 14, 16

5. Median of 1, 3, 5

6. Median of 4, 8, 12

7. Median of 2, 4, 6, 8, 10

8. Median of 3, 7, 9, 11

UNIT 6 | Worksheet continued

9. Mode of 2, 3, 3, 5

10. Mode of 4, 4, 6, 7

11. Mode of 1, 2, 2, 2, 3

12. Mode of 5, 6, 6, 7, 7, 7

13. Range of 2, 8, 11

14. Range of 4, 12, 15

15. Range of 7, 9, 20, 22

16. Range of 1, 6, 9, 16

UNIT 6 | Worksheet continued

17. P(heads) on fair coin

18. P(tails) on fair coin

19. P(rolling 1 on fair die)

20. P(rolling even on fair die)

21. Bag: 2 red, 3 blue. P(red)

22. Bag: 4 green, 6 yellow. P(green)

23. Bag: 1 black, 9 white. P(white)

24. Expected heads in 20 fair coin flips

UNIT 6 | Review quiz

Complete without help. 1 mark per question. Score: _____ / 4

1. Survey results: cats 8, dogs 12. How many students?

2. What fraction chose dogs?

3. A spinner has 4 equal sections, one red. P(red)?

4. What is the range of 14, 7, 21, 11?

What I found easy: _____

What I need to practise: _____

PARENT ANSWER KEY

Keep these pages separate from student copies. Equivalent fractions, units and mathematically equivalent answers are acceptable.

Unit 1 - Whole Numbers, Factors and Order of Operations

1. 6047 2. 9495 3. 21012 4. 66041 5. 4524 6. 6136

7. 33155 8. 34216 9. 336 10. 816 11. 360 12. 2912

13. 108 14. 105 15. 56 16. 108 17. 28 18. 50

19. 71 20. 81 21. 52 22. 63 23. 150 24. 150

Review quiz: 1. 1, 2, 3, 4, 6, 8, 12, 24 | 2. 7, 14, 21, 28, 35 | 3. 69,000 | 4. 432

Unit 2 - Fractions, Decimals and Percentages

1. 1/2 2. 2/3 3. 3/5 4. 3/4 5. 3/4 6. 3/5

7. 1/2 8. 2/3 9. 3/4 10. 5/6 11. 5/8 12. 1/2

13. 0.85 14. 6.55 15. 4.65 16. 10 17. 0.5 18. 0.75

19. 20% 20. 35% 21. 25 22. 30 23. 43 24. 15

Review quiz: 1. $45 | 2. 0.7 | 3. 9/20 | 4. 1/4 cup

Unit 3 - Patterns, Algebra and Equations

1. 15 2. 27 3. 32 4. 60 5. Add 5 6. Add 4

7. Subtract 5 8. Multiply by 3 9. 12 10. 21 11. 13 12. 13

13. 9 14. 18 15. 23 16. 17 17. 7 18. 9

19. 35 20. 36 21. 30 22. 32 23. 13 24. 23

Review quiz: 1. 23 | 2. 26, 31 | 3. n + 7 | 4. $5

PARENT ANSWER KEY (continued)

Unit 4 - Measurement, Area, Perimeter and Volume

1. 300 cm 2. 2.5 m 3. 4000 m 4. 1.5 km 5. 2000 mL 6. 0.75 L

7. 3000 g 8. 1.25 kg 9. 20 cm 10. 38 m 11. 36 cm 12. 60 m

13. 42 square cm 14. 42 square m 15. 64 square cm 16. 121 square m 17. 30 cubic cm 18. 64 cubic cm

19. 60 cubic cm 20. 60 cubic cm 21. 1 h 30 min 22. 135 min 23. 1 h 45 min 24. 1 h 25 min

Review quiz: 1. 18 m | 2. 20 square m | 3. 2500 mL | 4. 120 cubic cm

Unit 5 - Geometry, Angles and Coordinates

1. Acute 2. Right 3. Obtuse 4. Straight 5. 110 degrees 6. 55 degrees

7. 132 degrees 8. 90 degrees 9. 60 degrees 10. 60 degrees 11. 60 degrees 12. 60 degrees

13. 6 14. 8 15. 360 degrees 16. 90 degrees 17. I 18. II

19. III 20. IV 21. (2, -5) 22. (-4, -1) 23. (4, 2) 24. (-2, -1)

Review quiz: 1. 75 degrees | 2. 4 | 3. (0, 0) | 4. 180 degrees

PARENT ANSWER KEY (continued)

Unit 6 - Data, Graphs and Probability

1. 4 2. 7 3. 7.5 4. 13 5. 3 6. 8

7. 6 8. 8 9. 3 10. 4 11. 2 12. 7

13. 9 14. 11 15. 15 16. 15 17. 1/2 18. 1/2

19. 1/6 20. 1/2 21. 2/5 22. 2/5 23. 9/10 24. 10

Review quiz: 1. 20 | 2. 3/5 | 3. 1/4 | 4. 14

End of workbook

Extension idea: ask each student to write two new questions per unit, solve them, and exchange them with their brother.